Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Foundation models for biology are increasingly probed with interpretability methods that identify correlational structure – an attention head that aligns with a known biological signal, a scoring proxy that correlates with real experimental labels – and this correlational structure is often treated as evidence that a targeted causal intervention (activation steering, attribution-guided editing) will work as intended. We test this assumption directly on a protein language model, ESM2-650M, using difference-of-means activation steering, a causal intervention already shown to work for one property (thermostability). Under a pre-registered six-criterion protocol testing five novel target properties, we find no positive relationship between proxy validity and causal steering effect: our best-validated proxy (binding affinity, $r = 0.80$ held-out, confirmed by a weight-shuffle null control) steers nothing, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated causal effect. An independent, component-level mechanistic-interpretability result shows the identical pattern one level down: attention heads ranked by correlation with real 3D protein contacts show near-zero relationship to their causal ablation effect, confirmed via a full 480-head sweep. We further show that a single evaluation seed can produce a misleading verdict on a specific robustness criterion even when the underlying causal effect is stable, and introduce a diagnostic – steering-vector cosine similarity across targets – that explains an otherwise-unexplained result as a geometric artifact of a different, validated causal direction rather than independent evidence. These findings argue for validating explanation methods against downstream causal interventions, not treating correlational salience as self-certifying.

1 Introduction

A foundation model's internal correlational structure – which attention heads track a known signal, which features predict a real label – is frequently used as the basis for a claimed causal intervention: ablate this head because it looks important, steer along this direction because it correlates with the target property. Whether that correlational salience actually predicts the intervention's causal effect is rarely tested directly. We test it on ESM2-650M, a protein language model, using difference-of-means activation steering [Huang et al., 2025], a method already validated to causally move a real biological property (thermostability) by adding a direction – built from the residual-stream activation difference between real high- and low-property sequence groups – to the model during generation.

We ask: does a scoring proxy's correlation with real experimental labels predict whether steering toward it causally works? Across five novel target properties under one pre-registered protocol, the

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Table 1: Five novel target properties. Proxy r is Pearson correlation against real labels on a held-out split, computed before any steering (causal-intervention) run.

Target	Dataset	Proxy r	Steering result	Verdict
Binding affinity	ProteinGym DMS (RASK/DARPin)	0.80	flat null, all α	KILL
Catalytic activity	DLKcat (cross-protein)	0.22	PASS, 3/3 seeds	**PASS**
Intrinsic disorder	DisProt	0.45	PASS, 3/3 direction; 2/3 robustness	**PASS***
Immunogenicity	IEDB (4-tier)	$\leq$0.10	not run	KILL (pre-run)
Expression yield	eSol	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1, 2, 4, 6, and 2 of 3 seeds on criterion 3 (§3.1).

answer is no, and not weakly so. The best-validated proxy in our study ($r = 0.80$ held-out, for binding affinity) produces a flat null causal effect at every tested intervention strength, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a clean, monotonic effect that replicates across three independent random seeds. This mirrors an independent, purely mechanistic-interpretability result in the same model family: Vig et al. [2021] identified an attention head whose attention pattern correlates 12.9× with real 3D residue contacts, explicitly noting this was never tested causally; we ablate it directly and find an effect statistically indistinguishable from a low-correlation control, confirmed by a full sweep of all 480 heads in the model. Two levels – individual components and entire target properties – one lesson: correlational salience, however striking, is evidence about a model's learned representations, not a certificate that an intervention built on it will causally succeed.

2 Method: A Pre-Registered Protocol for Causal Validation

Before testing any target property, we lock a six-criterion decision rule, applied uniformly and computed automatically: (1) the real steering direction significantly beats a matched-norm random-direction control via paired bootstrap (10,000 resamples), not two separate baseline comparisons; (2) a coherent, monotonic dose-response across at least three steering strengths, restricted to a pre-established safe operating range (§4); (3) the effect survives a residue-exclusion robustness check ruling out pure compositional collapse; (4) the scoring proxy is validated against real experimental labels *before* the steering run – the causal-structure analogue of validating an explanation method against ground truth before trusting its downstream use; (5) the result beats the best existing technique where one exists; (6) at least 150 held-out evaluation sequences. A target failing any criterion outright is a **KILL**; one passing 1–2 but failing 3 or 4 is **AMBIGUOUS**.

3 Results: Explanatory Validity Does Not Predict Causal Success

Binding affinity. Using a ProteinGym deep-mutational-scanning assay of KRAS binding to a DARPin binder, we build a proxy from per-position mutational-sensitivity weights learned on the training split. This is the most rigorously validated explanation in our study: $r = 0.80$ held-out, confirmed by a weight-shuffle null control ($r = -0.001$, ruling out that the signal is generic rather than position-specific), held-out-position generalization ($r = 0.54$ on positions never seen during weight-fitting), and mutational-load extrapolation from single to double mutants ($r = 0.70$). The causal intervention built from this explanation is a flat, unambiguous null at every tested strength, with zero degenerate sequences – ruling out generation collapse as an alternative account. A plausible mechanistic reason (§4): this dataset's vector-building groups are single-backbone point mutants, giving the difference-of-means construction very little raw compositional signal to act on, independent of whether the underlying explanation is valid.

Catalytic activity. A glycine-minus-arginine compositional proxy on DLKcat (cross-protein enzyme turnover data), the weakest explanation of the four runnable targets ($r = 0.22$), matching the published activity-stability tradeoff in enzymology. This produces a clean, monotonic causal effect (Figure 2, left), significant at every safe-range α, surviving residue-exclusion, replicated identically across three independent seeds – the one target in our sweep with fully unanimous seed-robustness despite the weakest explanatory grounding.

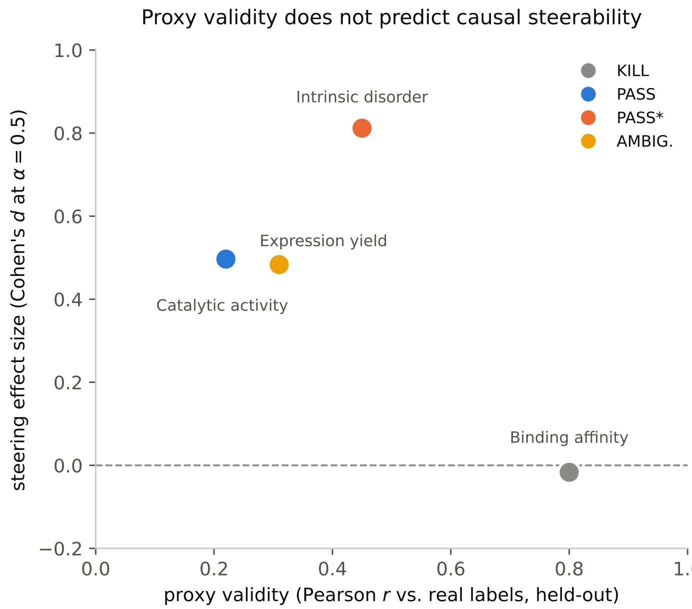

Figure 1: The four runnable targets, plotted as explanatory validity (proxy Pearson r against real labels) against causal steering effect size (Cohen's d at $\alpha = 0.5$). The most trustworthy explanation (binding affinity, $r = 0.80$) yields the least causally useful intervention; a far weaker explanation ($r = 0.22$, catalytic activity) yields one of the two most causally effective interventions.

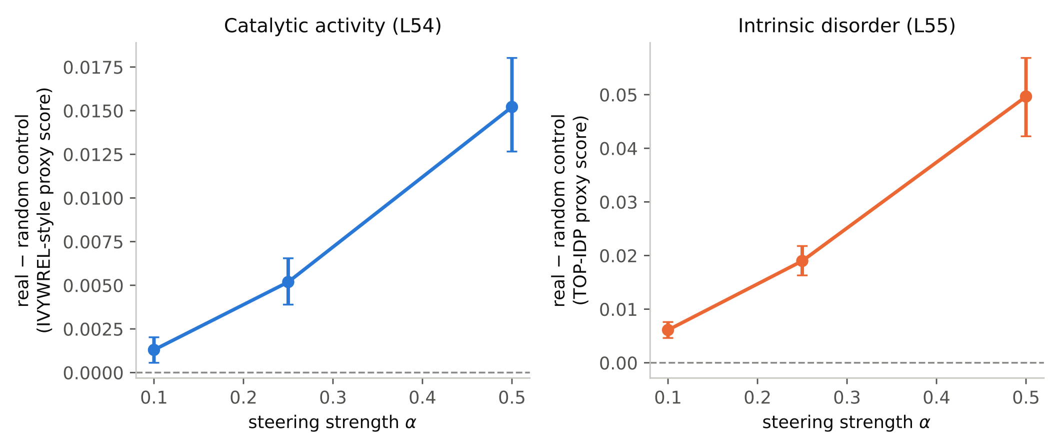

Figure 2: Dose-response for the two targets passing criterion 2. Both show a clean, monotonic increase with 95% bootstrap CIs bounded away from zero at every point.

3.1 Intrinsic disorder: seed-sensitivity in a robustness criterion

A proxy from the published TOP-IDP disorder-propensity scale [Campen et al., 2008] validates at $r = 0.45$ (the strongest explanation in our sweep), confirmed at the per-residue level (AUC 0.71 via a sliding window against real per-residue labels) – a case where explanatory validity and evaluation rigor are both high. The causal effect is significant and monotonically dose-responsive on all three independent seeds (Figure 2, right). However, the residue-exclusion robustness check – the step analogous to a faithfulness check on an explanation method – passes on 2 of 3 seeds (36–42% of effect magnitude retained, CI excludes zero) and fails on the third (10% retained, CI crosses zero, Figure 3). The causal effect's existence and direction are seed-stable; the specific robustness criterion is not, uniformly.

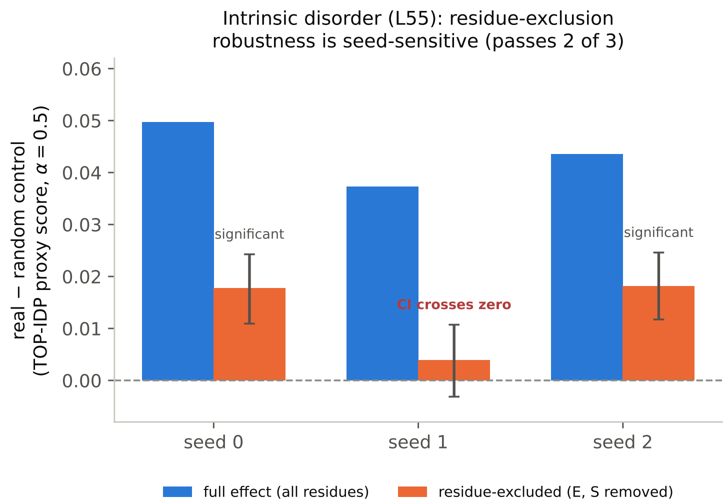

Figure 3: The intrinsic-disorder target's causal effect (blue) and the same effect after the two dominant substituted residues are excluded (orange, 95% bootstrap CIs), across three independent seeds. The underlying causal effect is stable across all three; the residue-exclusion faithfulness check is significant on seeds 0 and 2 but its CI crosses zero on seed 1 – the same causal effect, the same robustness criterion, three different verdicts.

Immunogenicity. We evaluate candidate explanatory proxies across four tiers of increasing fidelity to the real target. Proxies predictive on a surrogate binding-affinity tier ($r = 0.37$–0.43) collapse to near-chance on the real T-cell-response endpoint ($r \leq 0.10$); a full-length-antigen proxy reaching $r = 0.38$ is a source-organism confound – organism identity alone explains 58% of the apparent signal, and the same proxy's correlation flips to -0.32 once organism identity is held out of cross-validation, a direct example of an explanation that looks physically grounded but is actually tracking an unrelated confound. No proxy survives criterion 4, so no causal steering run was performed.

Expression yield. A net-charge-deviation proxy validates at $r = 0.31$–0.34 and produces a significant, dose-responsive causal effect that collapses under residue-exclusion – an explanation-validity check that passes but a faithfulness check that fails.

4 Grounding an Ambiguous Result: A Vector-Geometry Diagnostic

Rather than discarding the expression-yield result as unexplained noise, we compute cosine similarity between its steering vector and every other target's, layer by layer. We find substantial overlap with the intrinsic-disorder target's vector: $+0.38$ on the full 33-layer concatenation, rising to $+0.56$–0.67 at the three deepest layers. This is consistent with the independently documented biology – disordered protein regions are known to reduce solubility and expression yield – and grounds the result causally: expression yield's apparent effect is best understood as a partial, noisier echo of disorder's already-validated direction, filtered through a proxy more prone to compositional collapse, not independent evidence of a sixth causally steerable property. This diagnostic (compare a new intervention's direction against every other validated direction in the same study) is a cheap, generalizable faithfulness check for any study running multiple related causal interventions.

A caught faithfulness-evaluation artifact. As a mechanistic follow-up on thermostability, we tested whether restricting steering to the 5 layers previously identified as causally necessary preserves the effect of steering all 33. A first evaluation selected its best-performing α from the full tested range, including $\alpha \geq 1.0$, and concluded the 5-layer subset matched or exceeded the full intervention. This was an artifact of the evaluation, not a real result: at that α, the 33-layer intervention had already fully collapsed into degenerate output (zero non-degenerate evaluation sequences), while the 5-layer subset had not yet collapsed – an evaluation methodology issue directly analogous to comparing two explanation methods at operating points where one has already failed. Restricting the comparison to a range where neither arm has collapsed and rerunning end to end gives the honest result: the 5-layer subset produces a real, residue-robust effect, retaining only 43–59% of the full effect size.

The binding-affinity mechanistic account (§3) is likewise grounded, not asserted: the mean per-residue compositional distance between this target's low/high vector-building groups is 0.003, versus 0.02–0.05 for the three cross-protein targets in our sweep – a directly measurable reason the causal construction had little signal to act on, independent of the explanation's own validity.

5 Related Work

Huang et al. [2025] introduce difference-of-means activation steering for protein language models and validate it causally on thermostability; we extend their intervention to five new properties under a pre-registered protocol and test, to our knowledge for the first time, whether explanatory (proxy) validity predicts causal success across multiple unrelated properties. Vig et al. [2021] identify the attention head we redo as a genuine causal ablation test, extended here to a full 480-head sweep with zero correlational information in the ranking. Adams et al. [2025] train sparse autoencoders on ESM-2's residual stream and show, via linear probing – a purely correlational, post-hoc attribution method – that known determinants of thermostability and subcellular localization are identifiable in SAE feature space; their one steering experiment is a coherence check (does steering change fewer residues than a random direction), not a validation against a real causal ground truth, and their discussion states plainly that "investigating how to steer the model in useful ways remains an open direction for future work." Our work answers a version of that question directly: whether an interpretability method's correlational validity, evaluated rigorously by the field's own standards, predicts the causal usefulness of an intervention built on it.

6 Discussion

What "causal" means here. Every result in this paper is causal about the model's activation space – a real intervention on the forward pass, evaluated against a matched-norm random-direction control and a residue-exclusion faithfulness check, on generated (not merely scored) sequences – not validated against a real biological assay. A PASS means "causally steers a validated compositional explanation," not "causally engineers the underlying biological property," a distinction that matters for any claim that an interpretability method's causal usefulness has been established.

Limitations. All experiments use a single foundation model (ESM2-650M) at a single scale; we did not test whether these findings hold at other scales, an open question directly relevant to whether explanatory-validity/causal-effect mismatches are a property of this specific model or foundation models more broadly. All proxies are compositional heuristics rather than learned attribution methods, chosen for interpretability and speed; a stronger, more sophisticated explanation method might show a different relationship to causal effect. Each target's final proxy was selected as the best of several candidates scored on the same held-out split its validation r is reported on (rather than a separate select/test split), inflating every reported proxy-r in the same direction – the absolute values should be read as upper bounds, though this does not obviously overturn the ordinal mismatch the paper's central claim rests on. We ran many significance tests overall without a multiple-comparisons correction, partially mitigated by requiring multi-alpha/multi-seed agreement for any PASS, but individual borderline criteria are close enough to the noise floor that this remains a real caveat. We checked seed-robustness for two of five targets; we did not run any structural or functional plausibility check on generated sequences, so a PASS does not rule out that causally-steered sequences, while scoring well on the proxy, are otherwise physically or biologically implausible.

7 Conclusion

Correlational validity – of an attention head's alignment with a known biological signal, or of a scoring proxy's correlation with real experimental labels – does not predict causal effect when a downstream intervention built on it is actually tested, at two different levels of the same foundation model class. For interpretability methods intended to guide scientific discovery or trustworthy intervention in biology, this argues for validating explanations against the causal interventions they are meant to support, not treating correlational or attribution-based salience as self-certifying evidence of causal usefulness.

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